

TYLER PROCTOR

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Education

Massachusetts Institute of Technology (MIT) — GPA: 4.4/5.0

Sep. 2023 – May 2027

BS in Electrical Engineering and Computer Science, Minor in Finance, Conc. in Philosophy

Cambridge, MA

Technical Skills

Programming Languages: Python, JavaScript, Java, C++, Dart, CSS, HTML, C, Assembly, SystemVerilog, Bluespec
Developer Tools: VS Code, Android Studio, GitHub, Node, GitHub Copilot, Cursor, MongoDB, Render, PyTorch, ROS
Relevant Coursework: Linear Algebra, Calculus, Math for Comp Sci, Electrical Circuit Model & Design Physical Systems, Computational Structures, Intro to Algorithms, Intro to Probability, Intro to Machine Learning, Digital Systems Lab, Dynamic System Modeling & Control Design, Robotics: Science & System, & Hardware Architecture for Deep Learning

Experience

MIT FutureTech Lab

Feb. 2026 – Present

Researcher

Cambridge, MA

- Fine-tuned the MolmoAct2 VLA on the LIBERO dataset and custom robotic demonstration data, leveraging LoRA and distributed Slurm cluster job arrays on the ORCD HPC cluster to achieve a 89.9% decrease in training action loss
- Packaged custom environment bundles to benchmark the VLA on Franka Panda arms using the LeRobot framework
- Investigated training data size & task complexity across leading generative models to quantify robotic data scaling laws

Google LLC

May 2025 – Aug. 2025

Associate Software Developer Intern

Mountain View, CA

- Migrated 3 critical test processors to a newer universal framework, ensuring continuity of testing for 5 voice features
- Converted 2 release-critical end-to-end performance tests to operate within the newer framework for the Fuchsia operating system, while uplifting common testing behavior to a base class to ease creation of future tests
- Produced additional functionality to allow the tests to run on infra, making them release gated automated tests

Google LLC

Jun. 2024 – Aug. 2024

Student Training in Engineering Program (STEP) Intern

Mountain View, CA

- Managed the full development lifecycle from design documentation and code implementation to testing and internal launch, providing essential bug resolution assistance to over 220 Google Smart Home Engineers
- Produced the first internal automated tool leveraging Gemini AI to assist in diagnosis for incoming bugs reported through a new service created within the Smart Home servers
- Oversaw iterative tuning and testing of the Gemini AI model to significantly improve the reliability and efficiency
- Developed comprehensive utility libraries with C++ that establish API calls to multiple services, including Gemini and internal log systems, for universal use by internal Home engineer teams

Projects

RSS Robotic Racecar

Feb. 2026 – Present

ROS 2, Motor Control Theory

MIT

- Implemented a robust wall-following algorithm utilizing filtered LiDAR scan data and a Proportional-Derivative (PD) controller to achieve stable perimeter navigation, coupled with a safety controller that executed emergency braking
- Architected a visual servoing pipeline for autonomous line following and parking using color segmentation and homography transformations to achieve high-accuracy parking maneuvers and stable navigation
- Validated autonomous navigation pipelines against complex simulated urban environments, executing path planning, hazard avoidance, and automated parking, and secured 3rd place class-wide in a competitive lane-following time trial

FPGA-Powered Lane Detecting Autonomous Vehicle

Oct. 2025 - Dec. 2025

Field-Programmable Gate Array, SystemVerilog, Python, Vivado, PulseView, Surfer

MIT

- Utilized SystemVerilog on AMD's Spartan-7 FPGA on a robotic vehicle chassis to implement an image processing pipeline for camera input and filtering incoming image data to highlight upcoming lane markings
- Implemented a connected-component labeling (CCL) algorithm in hardware to mark and locate those filtered lanes

scrapTrip Website Project

Jan. 2024

Javascript, HTML, CSS, React, MongoDB, Render

MIT

- Created and deployed a social media website from scratch within the MIT class Web.lab over a period of two weeks
- Implemented a planning assistant system that utilized Mobi's Travel Planning API to give users feasible travel plans
- Tested live with over 30 real users to ensure usability

Extracurricular Activities

MIT Arcturus Club

Feb. 2024 – Jan. 2026

Autonomous Team Member

MIT

- Upgraded the computer vision system to improve color recognition for buoys and optimize autonomous navigation
- Developed a critical ROS2 connection system between Nvidia Jetson and Raspberry Pi, significantly enhancing communication stability and responsiveness for robot controls and navigation